Group II, Row IV: extraction of the BFKL and finite terms from the left-over expression

1 The left-over expression and its check

After the JIMWLK term and the DGLAP + RC terms have been extracted, the left-over is

$$\begin{aligned}\frac{d^3N_{\rm left}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]\\ &\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\\ &+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\ &\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,. \qquad (1)\end{aligned}$$

Write $C\equiv\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}$ (both prefactors of (1) equal C, since $\frac{g^2}{4\pi^2}\cdot\frac{g^2}{2\pi^2}N_c=\frac{g^4N_c}{8\pi^4}$), $L_\epsilon=\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}$, and $\bar\xi=\frac{k^+}{\vee}$.

The left-over is correct, and its two terms are identified as follows.

(a) The first term is the *endpoint piece* $-C\,h(1)\,\log(1-\bar\xi)$ of the soft channel, $h(1)=\big[\frac{d_\perp}{2}L_\epsilon+1\big]e^{-ik(y'-y)}=L_\epsilon\,e^{-ik(y'-y)}$: the full endpoint logarithm was $\log\frac{(\vee-k^+)k^+}{\vee\Lambda}=\log\frac{k^+}{\Lambda}+\log(1-\bar\xi)$, the $\log\frac{k^+}{\Lambda}$ part left with the JIMWLK term, and $\log(1-\bar\xi)$ stayed. (Checked numerically: soft channel minus JIMWLK minus the locally subtracted integral equals $-C\,h(1)\log(1-\bar\xi)$ to $O(\Lambda)$.)

Convention remark. This term is present because the DGLAP term was extracted in the *locally subtracted* plus-prescription convention, $\int_{\bar\xi}^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi}$. If instead the *standard* plus distribution on $[0,1]$ is used in $\mathcal{P}_{gg}$ — the convention in which $\mathcal{P}_{gg,+}=2N_c\big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\big]$ is the textbook splitting function — then $\int_{\bar\xi}^1\frac{h}{(1-\xi)_+}=\int_{\bar\xi}^1\frac{h-h(1)}{1-\xi}+h(1)\log(1-\bar\xi)$ absorbs this first term into the DGLAP integral, and it disappears from the finite block. The two bookkeepings differ by exactly this one term; nothing else changes.

(b) The second term is the *azimuthal* term: the L_ϵ-free part of the regrouped expression, $+C\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi}\big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-d_\perp\frac{(\mathcal{K}\cdot r)(\mathcal{K}'\cdot r)}{r^2}\big]f(\xi)$, $r=y'-y$, with $d_\perp\to2$ (legitimate, since no pole multiplies it) and the upper limit already extended to 1 at a cost $O(\Lambda)$.

2 Step 1: no rapidity logarithm hides in the left-over

Before anything can be "extracted", note that *neither term of* (1) *contains a* $\log\frac{\vee}{k^+}$:

(i) The coefficient of the first term is $\log(1-\frac{k^+}{\vee})$, which tends to 0 as $\vee\to\infty$ at fixed k^+. It does not grow with the rapidity interval; it is a finite endpoint (convention) term, not a rapidity logarithm.

(ii) The second term is finite as $\bar\xi\to0$. Two mechanisms protect it. First, at fixed positions, substituting $u=\frac{1}{\xi}$ gives

$$\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi}\,e^{-ik\cdot r/\xi}=\int_1^{1/\bar\xi}du\,\Big(\frac{1}{u}-\frac{1}{u^2}\Big)\,e^{-ik\cdot r\,u}\,, \qquad (2)$$

which *converges* as $\frac{1}{\bar\xi}=\frac{\vee}{k^+}\to\infty$ for $r\neq0$: the phase oscillates and the integral approaches a finite limit (checked numerically: the value stabilizes while $\log\frac{\vee}{k^+}$ would have grown without bound). Second, the dangerous collapse

region $r = O(\xi)$, where the phase stops oscillating, is doubly suppressed: writing $y' = y + \xi s$ costs a factor ξ^2 from the measure, and the azimuthal bracket has *exactly zero* angular average over the direction $\hat{r}$,

$$\int_0^{2\pi} \frac{d\theta_{\hat{r}}}{2\pi} \Big[\mathcal{K}\cdot\mathcal{K}' - 2\,(\mathcal{K}\cdot\hat{r})(\mathcal{K}'\cdot\hat{r}) \Big] = \mathcal{K}\cdot\mathcal{K}' - \mathcal{K}\cdot\mathcal{K}' = 0 \tag{3}$$

(in two dimensions $\langle 2\hat{r}^i\hat{r}^j\rangle_\theta = \delta^{ij}$).

Conclusion of Step 1. The BFKL logarithm cannot be found *inside* (1). It lives in the already-extracted DGLAP term: the soft channel $2N_c\frac{1-\xi}{\xi}\otimes L_\epsilon$ produces a hidden $\log\frac{\vee}{k^+}$ at its $\xi\to 0$ endpoint ($\xi\to 0$ is $p^+\to\vee$: the unmeasured gluon carries almost all of the projectile momentum and the measured gluon is the soft one — the projectile-side rapidity region), and a rapidity logarithm multiplying L_ϵ is not physical. The BFKL term is therefore produced by an *exact add-and-subtract*: a subtraction term with the same $\xi\to 0$ limit is added to the finite block (where it protects the DGLAP soft channel), and its exactly computed value is subtracted back. Adding zero changes nothing, so the construction may equally well be carried out on the left-over (1), which is what we now do.

3 Step 2: the subtraction term

Define, per ξ,

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\;\frac{1}{\pi}\,K(y,y',z)\,K(x,x',z)\;\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,, \tag{4}$$

with $K(a,b,z)=\frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}$, and the full subtraction

$$S \;\equiv\; -\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{d^3N_\xi^{\rm sub}}{d^3k} \;=\; -\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik(y'-y)\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)- \tag{5}$$

This is the real-emission structure with the parent at the common point z: the two soft-emission (dipole) kernels $K(y,y',z)K(x,x',z)$ of amplitude and conjugate, and the Wilson-line factor at coincident points, $\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U$ $W(z,z)$. It is IR and UV finite in $d=2$ and contains no μ. Its defining property: it has the *same* $\xi\to 0$ limit as the soft channel of the DGLAP term. Indeed, with the dimensionally regularized identity $\int_z^{(d=2-\epsilon)} K(y,y',z) = -\pi L_\epsilon$, the combination

$$2N_c\frac{1-\xi}{\xi}\Big[L_\epsilon\,\frac{d^3N_{\xi\,\rm LO}}{d^3k}+\frac{d^3N_\xi^{\rm sub}}{d^3k}\Big] = -\frac{2N_c(1-\xi)}{\xi}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int e^{-ik(y'-y)\frac{1}{\xi}}\,\frac{K(y,y',z)}{\pi}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')-K(x,z \tag{6}$$

has, after the exact change of variables $y = z+\xi(w-z)$, $y' = z+\xi(w'-z)$, a square bracket that vanishes *linearly* in ξ at fixed (w,w',z) (because $y,y'\to z$ and $\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) = K(x,x',z)$), and a z-integral that converges at large z in the new variables; so the combination is free of $\frac{2}{\epsilon}$, γ_E and μ, and its soft region gives $\int\frac{d\xi}{\xi}\times O(\xi) = \text{finite}$ — no $\log\frac{\vee}{k^+}$. This is why $+S$, once placed in the finite block, renders the extracted DGLAP soft channel harmless.

4 Step 3: the exact value of the subtraction

Eq. (4) depends on ξ only through the phase, so the same substitution evaluates S in closed form. At fixed z, set

$$y = z+\xi\,(w-z)\,,\qquad y' = z+\xi\,(w'-z)\qquad\text{(exact, not an approximation)}\,, \tag{7}$$

under which

$$d^2y\,d^2y' = \xi^4\,d^2w\,d^2w'\,,\qquad K(y,y',z) = \frac{K(w,w',z)}{\xi^2}\,,\qquad e^{-ik(y'-y)\frac{1}{\xi}} = e^{-ik(w'-w)}\quad\text{(exactly, since } y'-y=\xi(w'-w))\,, \tag{8}$$

and $W(z,z)$, $K(x,x',z)$ do not transform. Hence $\frac{d^3N_\xi^{\rm sub}}{d^3k} = \xi^2\times(\xi\text{-independent integral})$: the Jacobian ξ^4 times the kernel rescaling ξ^{-2}. The ξ^2 cancels the explicit $\frac{1}{\xi^2}$ of (5), leaving the elementary integral

$$\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi} = \Big[\log\xi-\xi\Big]_{\xi=\bar\xi}^{\xi=1} = -\log\bar\xi-(1-\bar\xi) = \log\frac{\vee}{k^+}-\Big(1-\frac{k^+}{\vee}\Big) \tag{9}$$

(upper minus lower; positive, as required for a positive integrand). Therefore

$$S = -\frac{1}{(2\pi)^3}\frac{g^4 N_c}{8\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)$$

(10)

5 Step 4: add and subtract, and the BFKL term

Insert $0 = S - S$ into (1), keeping S in its integral form (5) and replacing $-S$ by minus the closed form (10). Since (10) carries an overall minus sign, $-S$ is *plus* $\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\big[\log\frac{\vee}{k^+} - (1-\frac{k^+}{\vee})\big]\int KKW(z,z)$. Its $\log\frac{\vee}{k^+}$ part is the **BFKL contribution of this row**,

$$\boxed{\frac{d^3N^{\rm BFKL}_{\rm G\text{-}II,\,R\,IV}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)}$$

(11)

moved to the BFKL file (it appears there verbatim). Its coefficient is manifestly UV finite and μ-independent, as a BFKL kernel must be. The $-(1-\frac{k^+}{\vee})$ part of $-S$ is the boundary term of the finite block.

6 Result: the finite terms

Collecting everything,

$$\frac{d^3N_{\rm left}}{d^3k} = \frac{d^3N^{\rm BFKL}}{d^3k}\Big|_{(11)} + \frac{d^3N^{\rm finite}}{d^3k}\,,$$

(12)

with

$$\frac{d^3N^{\rm finite}}{d^3k} = \underbrace{-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,L_\epsilon\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,}_{\text{endpoint (plus-prescription convention) term}}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{(1-\xi)}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x}_{\text{soft subtraction } (=\,\mathrm{S})}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\Big(1-\frac{k^+}{\vee}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x'}_{\text{boundary}}$$

$$\underbrace{+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') - 2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]}_{\text{azimuthal}}$$

(13)

Bookkeeping

(i) (soft subtraction S) + (boundary) + (BFKL entry (11)) = 0 identically: the construction is an exact add-and-subtract of (10), and all three objects are UV finite.

(ii) The role of S in the finite block is to pair with the soft channel $2N_c\frac{1-\xi}{\xi}\otimes L_\epsilon$ of the *already extracted* DGLAP term: per ξ, their sum is the μ-free combination (6), whose bracket is $O(\xi)$, so the pair contains no $\log\frac{\vee}{k^+}\times L_\epsilon$. The DGLAP term alone and S alone each contain that unphysical double structure; only their placement in different named blocks makes it look otherwise.

(iii) The endpoint term (first entry of (13)) is the difference between the two plus-prescription conventions. If $\mathcal{P}_{gg}$ is used with the standard $[0,1]$ plus distribution, this entry is absorbed into the DGLAP term and disappears from (13); nothing else changes.

(iv) (BFKL (11)) + (13) reproduces the left-over (1) identically — Steps 2–4 only added zero.